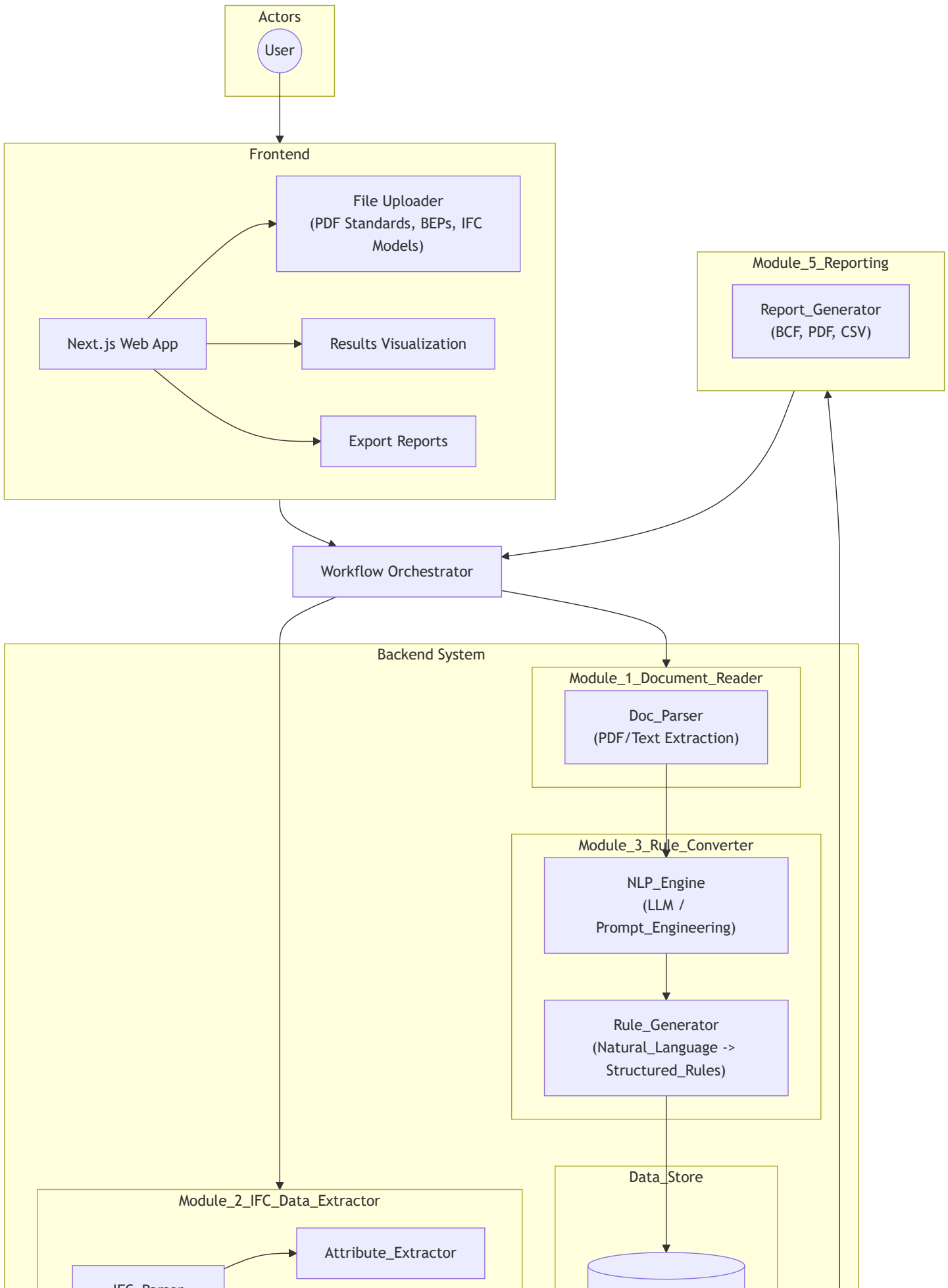


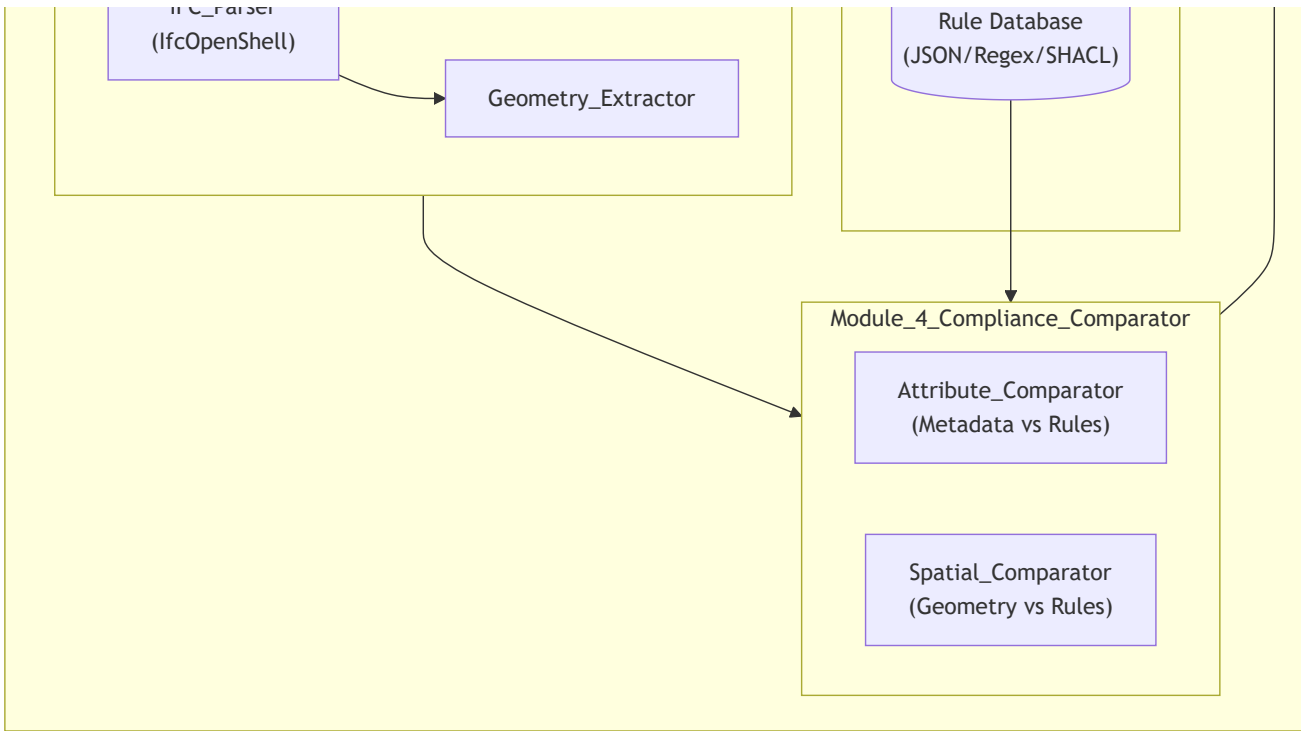
BIMGuard AI System Architecture

This document visualizes the system architecture for **BIMGuard AI**. It is structured around five core functional modules that automate the transition from static requirements to digital compliance.

1. High-Level System Architecture

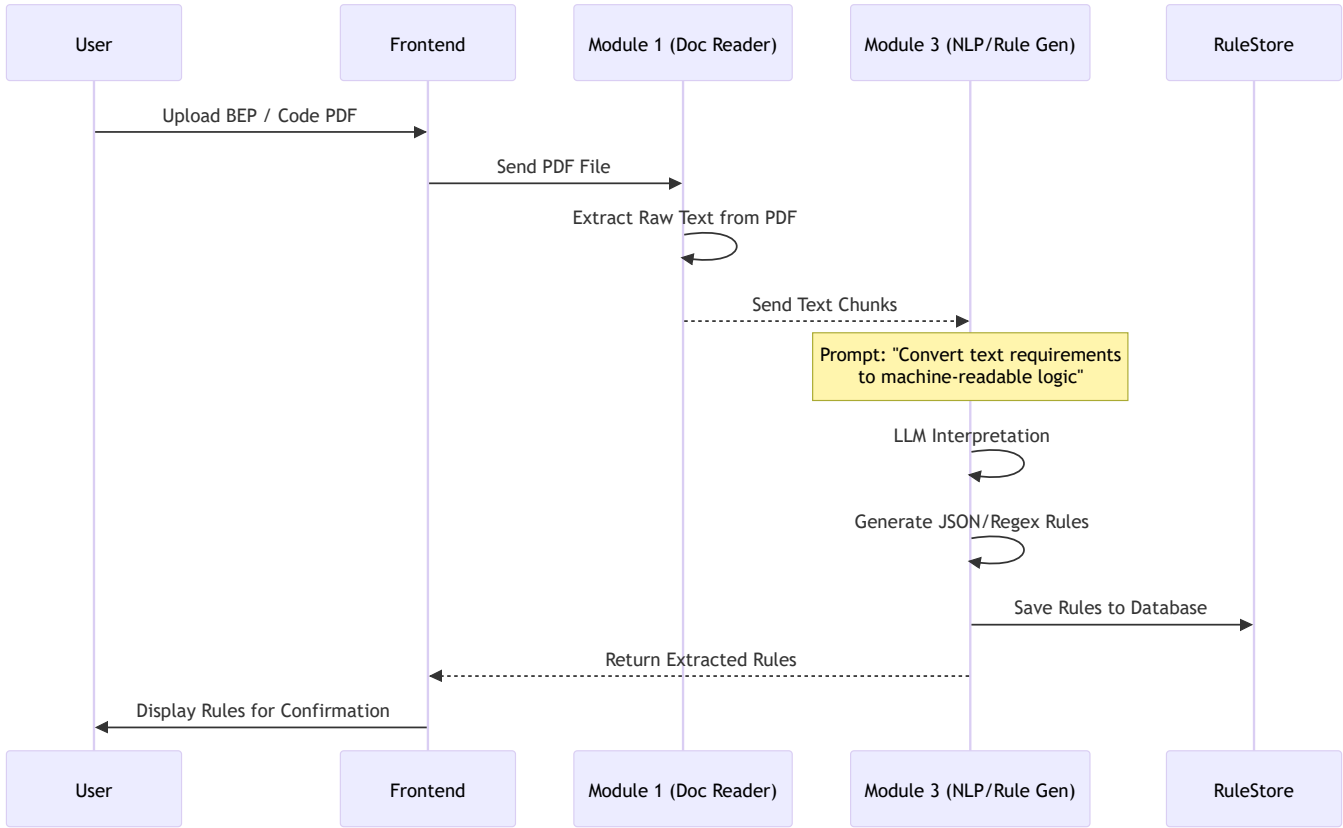
This diagram illustrates the system flow, categorized into the five requested modules: **(1) Document Reader**, **(2) IFC Reader**, **(3) Rule Converter**, **(4) Comparator**, and **(5) Reporting**.





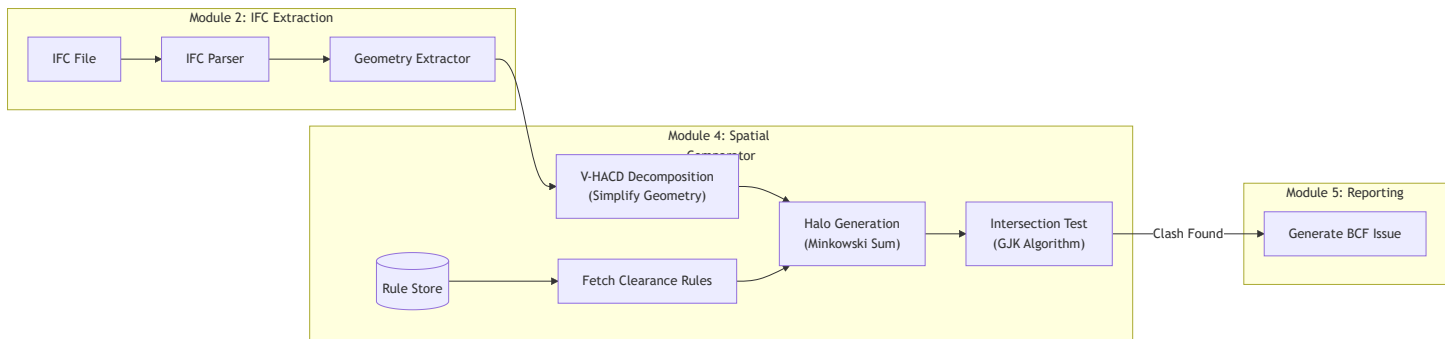
2. Module 1 & 3: Rule Extraction Workflow (Sequence)

This sequence details how the system reads documents (Module 1) and converts them into rules (Module 3).



3. Module 2 & 4: Compliance Checking Flow (Geometry)

This flow details the geometric analysis, where IFC data (Module 2) is checked against rules (Module 4).



4. Implementation Stack (Class Structure)

Below is the recommended codebase structure, organized by the five modules.

